

# R2 multi-scale bookkeeping: the power-sum lemma closes the N-counting; the D-loss reduces to one decoupling invocation

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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## 1. Purpose and scope

The arbitrary- $Q$  DR-2 conditional chain (dr2-cross-scale-induction v1.1) rests on two cited-not-reproduced residuals: R1 (local-to-global), now a written proof (R-034, dr2-t030-frontier-consolidation v1.1), and R2 (multi-scale bookkeeping). This note attacks R2. **Scope:** the bookkeeping of the one-step inequality

$$E_+(Q) \leq D^2 \left( \sum_{\theta} E_+(Q_{\theta})^{1/2} \right)^2$$

iterated across scales, where the  $\delta^{1/2}$ -caps  $\theta$  partition  $Q$ ,  $D$  is the  $\ell^2$ -decoupling constant at one scale, and (R-023, affine invariance) each  $Q_{\theta}$  rescales loss-free to a unit paraboloid patch. **Not claimed:** an unconditional  $N^{2+\varepsilon}$ ; any flip.

## 2. Pre-registered gates

G1: if the per-generation losses multiply unboundedly and no summable- $\varepsilon$  schedule or one-shot replacement removes the geometry dependence, record an HONEST PARTIAL with the residual named precisely. G2: success = a complete proof, every constant chase explicit, graded T6 PROVED CONDITIONAL on Bourgain–Demeter ONLY. **Outcome: G1.**

## 3. The N-counting is loss-free (power-sum lemma, PROVED)

**Lemma 1** (Power-sum / superadditivity). *For  $p \geq 1$  and nonnegative integers  $N_{\theta}$  with  $\sum_{\theta} N_{\theta} = N$ ,*

$$\sum_{\theta} N_{\theta}^p \leq \left( \sum_{\theta} N_{\theta} \right)^p = N^p,$$

*with equality iff at most one  $N_{\theta}$  is nonzero.*

*Proof.*  $x \mapsto x^p$  is superadditive on  $[0, \infty)$  for  $p \geq 1$ :  $(a + b)^p \geq a^p + b^p$  (convexity +  $0^p = 0$ ); induct on the number of parts. Strictness unless one term carries all the mass.  $\square$

**Consequence.** Substituting the induction hypothesis  $E_+(Q_{\theta}) \leq C N_{\theta}^{2+\varepsilon}$  into the one-step inequality,

$$E_+(Q) \leq D^2 \left( \sum_{\theta} (C N_{\theta}^{2+\varepsilon})^{1/2} \right)^2 = D^2 C \left( \sum_{\theta} N_{\theta}^{1+\varepsilon/2} \right)^2 \leq D^2 C (N^{1+\varepsilon/2})^2 = D^2 C N^{2+\varepsilon},$$

the last step by the Lemma with  $p = 1 + \varepsilon/2 \geq 1$ . The  $N$ -dependence is therefore carried with NO loss at ANY split, balanced or not – the maximally-unbalanced partition  $(N - 1, 1)$  obeys the Lemma exactly (machine: assert 1, 400 random partitions plus  $(N - 1, 1)$  for  $N \leq 499$ , worst lhs/rhs  $\leq 1$ ). This DISSOLVES the unbalanced-tree fear for the  $N$ -counting: the only thing that can blow up is the  $D$ -factor.

## 4. The D-counting: naive tree fails; submultiplicativity repairs

**The obstruction (naive fixed-scale tree).** Recursing the one-step inequality at a FIXED scale  $\delta_0$  multiplies a fixed  $D_0$  per generation. The maximally-unbalanced tree (peel one point per split) has depth  $G = N - 1$ , so the naive bound carries  $D_0^{2G} = D_0^{2(N-1)}$  – EXPONENTIAL in  $N$ , not  $N^{O(\varepsilon)}$  (machine: assert 3,  $N = 30$  gives  $D_0^{2 \cdot 29} \approx 2.9 \times 10^{17} = e^{40.2}$ ). Any R2 proof that silently assumes balanced splits or depth  $O(\log N)$  is therefore WRONG; this is the documented failure mode.

**The repair (submultiplicativity / one-shot).** The  $\ell^2$ -decoupling constant is submultiplicative across scales,  $D(\delta_1 \delta_2) \leq C_{\text{sub}} D(\delta_1) D(\delta_2)$ , and the Bourgain–Demeter theorem packages the entire scale-1-to- $\delta$  passage as a SINGLE application with a SINGLE loss  $D(\delta) \leq_\varepsilon \delta^{-\varepsilon}$  (the multi-scale induction is INTERNAL to its proof). Decoupling directly from scale 1 to the finest relevant scale  $s$  (the min separation, where every cap holds  $\leq 1$  point = base case) gives

$$E_+(Q) \leq_\varepsilon s^{-2\varepsilon} N^2,$$

ONE loss, independent of tree shape. For polynomially-separated  $Q$  ( $s \geq N^{-C}$ ) this is  $s^{-2\varepsilon} \leq N^{2C\varepsilon} = N^{O(\varepsilon)}$  (machine: assert 4,  $s = N^{-3}$  gives the fixed power  $N^{6\varepsilon}$ , log-dominating the exponential naive factor at  $N = 1000$ : 4.1 vs 1384.9).

## 5. The residual, named precisely (gate G1)

For ARBITRARY (not polynomially-separated)  $Q$ , the min separation  $s$  can be sub-polynomial, and then  $s^{-2\varepsilon}$  is NOT  $N^{O(\varepsilon)}$ : the adversarial  $s = e^{-N}$  gives one-shot exponent  $2\varepsilon N / \log N$ , which GROWS unboundedly ( $1.8 \rightarrow 10.5 \rightarrow 74.9$  at  $N = 30/300/3000$ ; machine: assert 5). So telescoping ALONE does not deliver the uniform-in-geometry bound. The single remaining ingredient is exactly:

[BD-DISCRETE]: the Bourgain–Demeter discrete well-separated reduction,

which pigeonholes the  $O(\log N)$  relevant scales and reduces the arbitrary configuration to the separated base case with a single  $N^{O(\varepsilon)}$  loss – equivalently, accepting polynomial separation  $s \geq N^{-O(1)}$  after the affine normalization. This is standard decoupling machinery (the SAME single  $\delta^{-\varepsilon}$  Bourgain–Demeter prove), invoked, not reproduced.

**Net effect on R2.** Before: R2 = “vague multi-scale bookkeeping over  $O(\log N)$  levels.” After: the  $N$ -counting is a PROVED loss-free power-sum lemma; the  $D$ -counting is a single [BD-DISCRETE] invocation. R2 is reduced to ONE named, standard ingredient. With R1 proved, the arbitrary- $Q$  chain rests on: the separated base case (T6 cond. on BD) + [BD-DISCRETE]. The chain’s honest grade is unchanged – **T6 PROVED CONDITIONAL on Bourgain–Demeter** – but its residual set is now a SINGLE clean citation.

## 6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- ( $\alpha$ ) “*The power-sum lemma is trivial – does it really carry the unbalanced-tree worry?*” VALID, and that is the point: the worry was MISPLACED on the  $N$ -counting (where it is dissolved by a one-line convexity fact) and BELONGS to the  $D$ -counting (assert 3), which submultiplicativity handles. Separating the two is the contribution; neither half is deep, but the conflation was what kept R2 vague.
- ( $\beta$ ) “*Submultiplicativity of the decoupling constant for  $C^2$  graph patches is itself cited.*” UPHELD as the residual: yes – that is precisely [BD-DISCRETE], named as the single remaining ingredient. The note does NOT claim to reprove Bourgain–Demeter; it claims to reduce R2 TO it, cleanly.

- ( $\gamma$ ) “Affine rescaling to a unit paraboloid patch might lose the curvature that makes decoupling work.”  
 DISMISSED: parabolic rescaling maps a paraboloid patch to a paraboloid patch EXACTLY (R-023; dr2-affine-invariance), so curvature is normalized, not lost – the flat-GAP escape (large planar additive energy) is illusory because the normal coordinates impose the curved constraint at every scale.

**Quantitative sanity check (mandatory).** The power-sum slack is exactly 0 at the trivial partition and strictly positive otherwise (assert 2: (100,100) slack 0.034); the naive vs one-shot comparison is in log scale to avoid the  $D_0^{2(N-1)}$  overflow ( $N=1000$ : 1384.9 vs 4.1); the adversarial exponent growth is monotone and unbounded (assert 5). All five checks PASS, exit 0; the verdict string in the JSON artefact records G1-PARTIAL.

## 7. Result footer

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Result ID:              | B5-BEYOND-LAYER-BOUND / DR-2 R2 multi-scale bookkeeping (T-030 conditional chain)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Precise statement:      | R2 PARTIAL ADVANCE (gate G1). N-counting: the power-sum lemma $\sum N_{\theta}^p \leq N^p$ ( $p \geq 1$ ) is PROVED, so the recursion's N-dependence is loss-free at any split balance (the unbalanced-tree fear is dissolved). D-counting: a naive fixed-scale tree accumulates $D^{2(N-1)}$ (exponential, demonstrated), but submultiplicativity collapses it to one critical-scale loss $s^{-2\epsilon} N^2$ . Residual NAMED: [BD-DISCRETE] (Bourgain-Demeter discrete well-separated reduction) removes the geometry-dependent $s^{-\epsilon}$ ; equivalently poly-separation $s \geq N^C$ . R2 reduced from vague bookkeeping to ONE clean invocation. |
| Scope:                  | the multi-scale bookkeeping residual only; with R1 (R-034) proved, the arbitrary-Q chain rests on separated base + [BD-DISCRETE].                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Evidence grade:         | PARTIAL ADVANCE; the chain stays T6 PROVED CONDITIONAL on Bourgain-Demeter (residual set tightened {R1,R2} $\rightarrow$ single [BD-DISCRETE]).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Reproduction command:   | python codes/vacuum/dr2_t030_r2_bookkeeping.py                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Falsifier:              | a partition with $\sum N_{\theta}^p > N^p$ ( $p \geq 1$ ); a fixed-scale tree bound below $D^{2G}$ on the unbalanced tree; a claim that telescoping alone gives $N^{2+\epsilon}$ for sub-polynomial $s$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Tier before / after:    | none -- operator decides. B5 stays T6-conditional; T-030 stays OPEN for unconditional $N^{2+\epsilon}$ ; arbitrary-Q chain stays T6 PROVED CONDITIONAL on BD.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| No-overclaim statement: | R2 is REDUCED, not closed; [BD-DISCRETE] is cited not reproduced; no unconditional bound; no flip.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Next required action:   | operator: register the R2-reduction result (ledger) if deemed reuse-worthy; decide whether the single-invocation chain warrants a DR2-SHARE / arbitrary-Q tier note. T-030 unconditional remains the open frontier.                                                                                                                                                                                                                                                                                                                                                                                                                                          |